

# Information Invariance: What It Is, What It Decides, and What It Does Not Derive

Monty Dabas<sup>1</sup>

<sup>1</sup>*Celextrix Private Limited, New Delhi 110086, India\**

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We define information invariance for thermodynamic response charts as the stationarity of the loop functional of the entropy-scaled accessibility one-form, and settle what follows from it. Three structural results are proved on finite charts with executable certificates: information invariance is equivalent to vanishing thermodynamic curvature and to exactness of the one-form, i.e. to the Onsager-reciprocal sector; microscopically it is exactly time-reversal invariance of the statistical weight, through the factor  $\chi = W(\Theta\Gamma)/W(\Gamma)$ ; and curvature is a sufficient but not a necessary witness of that invariance, since microscopic asymmetry can cancel under coarse-graining. On the experimental side we prove that the loop area of a *single* thermal cycle decides nothing — a flat chart with a one-time irreversible offset and a curved chart give equal first-cycle areas — which we illustrate on our own graphene Raman data, where the measured loop does not close. We then carry every obligation of the associated emergent-dynamics programme to a verdict rather than leaving it open: the quadratic energy law is not derivable from invariance but is uniquely selected by a stated minimal-scale principle; the linearised dynamics is a damped rotation, not a Schrödinger generator, and we give the exact condition that would make it one; the Planck length is an input rather than a consequence, the scale the theory supplies being cosmological; the emergent-Einstein claim fails on degree-of-freedom counting, on the  $\hbar$ -dependence of the thermodynamic route, and on non-conservation of the scalar source; and the proposed decoherence rate is dimensionally invalid, its unique repair being the Diósi–Penrose expression. We regard the negative results as the paper’s main contribution: they replace an unsupported programme with a small set of proved statements and explicitly stated conditions.

## I. INTRODUCTION

Onsager reciprocity is derived under microscopic reversibility. Entropy-weighted reparameterisations of response functions yield an effective coupling matrix whose cross-coefficients are generically asymmetric [1]; the associated geometric statement is that the accessibility one-form  $\omega = \lambda_p dp + \lambda_v dv$  need not be closed. This raises a structural question: *which* irreversible trajectories are admissible.

The present paper separates what can be proved from what is presently conjectured. Section II defines information invariance as a loop functional. Section III proves two finite theorems about it, with machine-checked certificates. Section V states the empirical situation honestly, including a non-closure finding in our own data that weakens the naive reading of loop area. Section VI states the dynamical programme as conjectures with explicit obligations.

## II. INFORMATION INVARIANCE, DEFINED

**Definition 1** (Information invariance). *Let  $\omega$  be the entropy-scaled accessibility one-form on a response chart. Define the loop functional  $\mathcal{I}(\gamma) = \oint_\gamma \omega$  for closed paths  $\gamma$ . The chart is information invariant if  $\mathcal{I}(\gamma) = 0$  for every admissible closed  $\gamma$ , written  $\delta\mathcal{I} = 0$ .*

This is a selection principle: it constrains admissible evolution, and asserts neither conservation of microscopic information nor invariance under arbitrary transformations. It postulates no new interaction and does not quantise spacetime.

## III. PROVED FINITE STRUCTURE

Discretise the chart on a lattice with rational edge values;  $\Omega = d\omega$  is the plaquette circulation, and discrete Stokes holds exactly.

**Theorem 1** (Information invariance = flat sector). *On a finite simply connected chart the following are equivalent: (i)  $\delta\mathcal{I} = 0$ ; (ii)  $\Omega \equiv 0$ ; (iii)  $\omega$  is exact, i.e. a potential  $\Phi$  exists and line integrals are path independent. On the  $\lambda$ -chart with  $\lambda_X = -ST/C_X$ , (ii) is the flat-closure sector  $\Gamma_c\Gamma_m = 1$ , i.e. Onsager reciprocity.*

*Proof sketch.* (ii) $\Rightarrow$ (i) by discrete Stokes; (i) $\Rightarrow$ (ii) by taking  $\gamma = \partial(\text{plaquette})$ ; (i) $\Leftrightarrow$ (iii) is path independence  $\Leftrightarrow$  exactness on a connected graph. All three directions, plus planted-curvature negative controls, are executed over exact rationals in the accompanying certificate.  $\square$

**Theorem 2** (One cycle does not witness curvature). *There exist a flat chart with a single irreversible offset and a curved chart without offset whose first-cycle loop areas are equal. The two are separated only by the second cycle: the offset construction collapses (reopen ratio 0), the curvature construction reopens with the same area (ratio 1).*

\* [montydabas@celextrix.org](mailto:montydabas@celextrix.org)

Consequently a nonzero loop area measured on one cycle is consistent with irreversible drift and does not by itself establish  $\iint \Omega \neq 0$ . Certificates for both theorems are in [papers/information-invariance/certificates](#); the pinned SHA-256 is reproduced by continuous integration on every commit.

#### IV. WHAT INFORMATION INVARIANCE MEANS MICROSCOPICALLY

The companion work derives, for an ensemble with weight  $W$  on a phase space carrying a time-reversal involution  $\Theta$ , the identity  $L_{ij} - \varepsilon_i \varepsilon_j L_{ji} = \int \langle (1 - \chi) J_i(0) J_j(t) \rangle_W dt$  with  $\chi(\Gamma) = W(\Theta\Gamma)/W(\Gamma)$ , and identifies the coarse-grained antisymmetric part with  $\Omega$  [1]. Chaining this with Theorem 1 gives information invariance a microscopic reading, and one leg of the chain turns out to be one-directional.

**Theorem 3** (Microscopic characterisation). *On a finite phase space, with  $C_W(f, g) = \sum_{\Gamma} W(\Gamma) f(\Gamma) g(\Theta\Gamma)$ ,*

$$C_W(f, g) - C_W(g, f) = \sum_{\Gamma} W(\Gamma) [1 - \chi(\Gamma)] f(\Gamma) g(\Theta\Gamma), \quad (1)$$

*exactly, and the following are equivalent:  $\chi \equiv 1$ ;  $C_W$  is symmetric for all observables; the antisymmetric response vanishes. Hence  $\delta\mathcal{I} = 0$  is exactly time-reversal invariance of the statistical weight.*

**Theorem 4** (Curvature is sufficient, not necessary).  $\chi \equiv 1 \Rightarrow \Omega = 0$  *always. The converse fails: for  $\Theta = (01)(23)$  and  $W = (3, 1, 1, 3)$  the two orbit contributions are  $+2$  and  $-2$ , so  $\chi \neq 1$  while the coarse-grained total vanishes.*

Together with Theorem 2 this bounds what a hysteresis experiment can decide:

$$\begin{aligned} \Omega \neq 0 \text{ across cycles} &\Rightarrow \chi \neq 1: \text{informative} \\ \Omega = 0 &\Rightarrow \text{nothing about } \chi \\ \text{one-cycle loop area} \neq 0 &\Rightarrow \text{nothing about } \Omega \end{aligned}$$

The predeclared second-cycle test is therefore not a precaution but the only step that carries an inference.

#### V. EMPIRICAL SITUATION

Temperature-cycled Raman measurements on monolayer CVD graphene on  $\text{SiO}_2$  [1] give reproducible first-cycle hysteresis in the  $G$  and  $2D$  peak positions, with loop areas well above fit uncertainty (independently refitted:  $744 \pm 22$  and  $692 \pm 28 \text{ cm}^{-1}\text{K}$  at spot 1). The loop, however, does *not* close at 300 K: the cooling branch returns  $+4.95 \text{ cm}^{-1}$  ( $G$ , spot 1) above the heating branch. By Theorem 2 this first-cycle observation is compatible with an irreversible component (adsorbate desorption or strain release) and does not establish reversible curvature. A predeclared decision rule has been registered:

reopen ratio  $\geq 0.6$  on both bands supports curvature;  $\leq 0.2$  indicates irreversible drift [2]. The measurement is pending.

#### VI. CONJECTURED DYNAMICS AND ITS OBLIGATIONS

The following are stated as conjectures. Each names the obligation a derivation must discharge; none is used to support the claims of Sec. III.

We take as the working dynamics, with  $u = ct - r$  and energy density  $\mathcal{E}(u, \theta, \phi)$ ,

$$\partial_u \mathcal{E} = \alpha_G \mathcal{E}^2 + \alpha_\Lambda \mathcal{E}_{\text{vac}} + \epsilon \mathcal{F}[\mathcal{E}], \quad (2)$$

and ask what, if anything, justifies it.

**Theorem 5** (Invariance alone selects nothing). *On an information-invariant chart  $\omega = d\Phi$ , and every potential-transporting evolution  $\partial_u \Phi = f(\Phi)$  preserves exactness, for arbitrary  $f$ . The dynamics compatible with  $\delta\mathcal{I} = 0$  therefore form an infinite-dimensional family, and (2) is not derivable from Definition 1.*

**Theorem 6** (Minimal scale usage selects the quadratic law). *For  $\partial_u \mathcal{E} = \alpha_n \mathcal{E}^n$  the coupling must satisfy  $[\alpha_n] = M^{1-n} L^{n-2} T^{2n-2}$ . No  $n$  admits a coupling built from  $\{G, c\}$  alone. Admitting a single new length  $\lambda$  gives uniquely  $\alpha_n = G^{n-1} c^{4-4n} \lambda^{2n-3}$ , and requiring  $\lambda$  to appear exactly once ( $d = 1$ ) forces  $n = 2$ , with  $\alpha_2 = G\lambda/c^4$ .*

Two consequences. First, the quadratic term is not arbitrary once one states the selection principle — one new scale, used once — but it does not follow from invariance. Second, the coupling written as  $\alpha_G \sim G/c^4$  is short by exactly one length; by Theorem 14 the length the theory itself supplies is the cosmological  $L_{\mathcal{E}}$ , giving  $\alpha_2 = 1/(c^2 \sqrt{G\mathcal{E}})$ . We record (2) as a conjecture with that stated provenance:

**Conjecture 1** (Null-coordinate realisation). *Equation (2) with  $\alpha_2 = G\lambda/c^4$  is the physically realised dynamics. **Obligation D1'**: justify the selection principle itself, and the identification  $\lambda = L_{\mathcal{E}}$ .*

#### A. Where a complex phase can come from (proved)

The step usually left implicit is the appearance of a complex phase in a real theory. Its finite content is settled.

**Theorem 7** (Real flow = Schrödinger flow). *Let  $x'$  =  $Ax$  be a real linear flow on  $\mathbb{R}^{2n}$  and  $J$  an orthogonal complex structure ( $J^2 = -I$ ,  $J^T = -J$ ). Then  $A^T = -A$  and  $AJ = JA$  if and only if  $H := -\hbar JA$  is Hermitian for  $J$ , in which case the flow reads  $i\hbar\psi' = H\psi$  under  $x \leftrightarrow \psi$ ; equivalently, the flow is complex linear and norm preserving.*

**Theorem 8** (One-channel obstruction). *An odd-dimensional real flow admits no complex structure, since  $\det(J)^2 = \det(-I) = (-1)^N$ . In particular a single scalar channel  $x' = ax$  is never a Schrödinger flow, for any  $a$  and any rescaling.*

Linearising Conjecture 1 about a background gives, at leading order, the single real channel  $\partial_u \delta \mathcal{E} = 2\alpha_G \bar{\mathcal{E}} \delta \mathcal{E} + \epsilon \mathcal{F}' \delta \mathcal{E}$ . By Theorem 8 this cannot be Schrödinger evolution as it stands, and by Theorem 7 what a derivation must supply is explicit:

**D2a** a conjugate channel  $\pi$  of  $\delta \mathcal{E}$  within the dynamics, and a complex structure  $J$  on the pair  $(\delta \mathcal{E}, \pi)$ ;

**D2b** antisymmetry of the linearised generator on that pair, i.e. the quadratic self-interaction must contribute to the Hermitian  $H$  and not to a symmetric part of  $A$ .

The first is supplied by structure that is already derived, not assumed. An oriented cut yields a quarter turn  $\iota$  with  $\iota^2 = -1$  and  $\iota^\dagger = -\iota$  [4], and every generator splits uniquely relative to the cut  $J$  into an even part preserving the two sheets and an odd part transporting between them [5]. Taking the conjugate channel to be the image on the opposite sheet gives:

**Theorem 9** (Cut supplies the pair; phase is the odd part). *Let  $J$  be a balanced cut, and write the generator in the sheet basis as blocks  $A = \begin{pmatrix} A_{++} & B \\ C & A_{--} \end{pmatrix}$  with  $\iota = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}$ . Then  $A\iota = \iota A$  iff  $A_{++} = A_{--}$  and  $C = -B$ ;  $A^T = -A$  iff the diagonal blocks are antisymmetric and  $C = -B^T$ ; both hold iff the diagonal blocks are equal and antisymmetric and  $B$  is symmetric, in which case  $H = -\hbar \iota A$  is Hermitian and Theorem 7 applies. A purely cut-even generator never mixes the sheets and produces no phase, and the symmetric part of  $A$  is exactly the damping:  $d|x|^2/du = 2x^T A_{\text{sym}} x$ .*

## B. The computation, and its negative outcome

Theorem 9 leaves one computation. We perform it on a transverse lattice with the parity cut, in exact arithmetic. The linearisation of Conjecture 1 about a background contributes three operator types: pointwise multiplication  $M$  by  $2\alpha_G \bar{\mathcal{E}}$  from the quadratic term, a second difference  $L$  from any diffusive part of  $\mathcal{F}$ , and a first difference  $D$  from any transport part. We find:

- $M$  is symmetric and cut even; it contributes  $d|x|^2/du = 4\alpha_G \bar{\mathcal{E}} |x|^2$ , i.e. pure growth or decay, and no phase;
- $L$  is symmetric and negative semidefinite: diffusion, no phase;
- $D$  is antisymmetric and cut odd: it alone preserves the norm and meets the conditions of Theorem 9;

- for  $A = M + L + D$  the symmetric part is exactly  $M + L$ , which does not vanish at a nonzero background; a coordinate rescaling is a similarity and cannot remove it.

**Result.** The linearised dynamics of Conjecture 1 is *not* a Schrödinger generator. It is a damped (or amplified) rotation. The emergent-quantum step therefore does not follow from the equation as written, and we withdraw it as a claim. What survives is a sharp conditional: the step holds if and only if the dynamics is amended so that the cut-even symmetric part cancels at the background,  $M + L = 0$ , leaving the cut-odd transport term as the generator. Theorems 1 and 2 are independent of this and are unaffected.

## C. Emergent geometry: three obstructions

**Theorem 10** (Degree-of-freedom no-go). *In four dimensions the metric has ten components; four coordinate conditions and four constraints leave two propagating polarisations. The dynamics carries one scalar function, so it can reproduce at most a one-function sector (spherically symmetric or conformally flat) and cannot generate two independent polarisations.*

**Theorem 11** (The Clausius route imports  $\hbar$ ). *No temperature is constructible from acceleration and  $c$  alone in  $(M, L, T, \Theta)$ ; the Unruh temperature requires  $\hbar$  and  $k_B$ . Hence the derivation of the field equations from  $dQ = T dS$  [6] is unavailable to a dynamics claiming to be deterministic and non-quantised.*

**Theorem 12** (The scalar source is not conserved). *The contracted Bianchi identity forces  $\nabla^\mu T_{\mu\nu} = 0$ . The law  $\partial_u \mathcal{E} = \alpha \mathcal{E}^2 + \dots$  is not a divergence: on a periodic transverse lattice the total changes, whereas a genuine divergence leaves it fixed.*

We therefore withdraw the emergent-gravity claim as a derivation. Three conditional routes remain and are stated rather than assumed: restrict to the one-function sector; admit  $\hbar$  explicitly and give up deriving quantum structure; or supply the compensating flux, or a second field, that restores conservation of the source.

## D. The regulator scale (settled)

The finiteness claim usually attached to Conjecture ?? is  $\langle \mathcal{E}(x) \mathcal{E}(x') \rangle \sim (r^2 + L_P^2)^{-1}$  with  $L_P = \sqrt{\hbar G/c^3}$ . Dimensional analysis decides its status.

**Theorem 13** (No Planck length without  $\hbar$ ). *In the  $(M, L, T)$  dimension group,  $\hbar$  is not in the span of  $\{c, G\}$ , and no product  $G^a c^b$  is a length. Hence  $L_P$  cannot be constructed from the deterministic constants; admitting it presupposes  $\hbar$ .*

**Theorem 14** (The theory’s own regulator is cosmological). *From  $\{c, G, \mathcal{E}\}$  a length does exist, uniquely up to powers:  $L_{\mathcal{E}} = c^2/\sqrt{G\mathcal{E}}$ , equivalently  $\Lambda^{-1/2}$ . For a vacuum energy density of the observed order,  $L_{\mathcal{E}}/L_P \sim 10^{61}$ .*

Consequently the model regularises, if at all, at a cosmological scale; inserting  $L_P$  is an assumption and the ultraviolet-finiteness claim of the earlier draft is withdrawn. The theory must either admit  $\hbar$  as a primitive — forfeiting the claim to derive quantum structure — or work with  $L_{\mathcal{E}}$ .

### E. The decoherence rate (corrected)

**Theorem 15** (The printed rate is not a rate).  *$Gm^2/(\hbar V)$  has dimensions  $M^0L^{-2}T^{-1}$ : a rate per unit area. The unique repair replacing the volume by a length,  $Gm^2/(\hbar R)$ , is dimensionally a rate and is exactly the Diósi–Penrose expression  $E_G/\hbar$  with  $E_G \sim Gm^2/R$  [7, 8].*

With  $m = 10^{-17}$  kg and  $R = V^{1/3} = 10^{-7}$  m the repaired rate is  $6.3 \times 10^{-4} \text{ s}^{-1}$ , of the order quoted in the earlier draft — but it is a Diósi–Penrose number, recovered only after the correction. We therefore state the result as a *consistency requirement*: any emergent gravity of this kind must reproduce  $E_G/\hbar$  in this regime. It becomes a prediction only if one of the following is supplied: a derivation of the coefficient, which DP fixes only through a choice of regularisation length; a mass or geometry regime of departure; or a dependence on shape or entropy gradient, which DP does not carry. Optomechanical tests discriminate such departures [9, 10], not the agreement.

## VII. SUMMARY OF STATUS

Every claim below is either proved with a machine-checked certificate that is regenerated on each commit, or marked as a stated condition. No claim rests on an argument that is not reproduced by the accompanying code.

| Statement                                                                       | Status              | W |
|---------------------------------------------------------------------------------|---------------------|---|
| $\delta\mathcal{I} = 0 \Leftrightarrow \Omega = 0 \Leftrightarrow \text{exact}$ | proved              | T |
| One cycle does not witness curvature                                            | proved              | T |
| Real flow is Schrödinger iff $J$ , antisym.                                     | proved              | T |
| One channel admits no complex structure                                         | proved              | T |
| Cut supplies the pair; phase is odd part                                        | proved              | T |
| No Planck length without $\hbar$                                                | proved              | T |
| Own regulator is cosmological                                                   | proved              | T |
| Printed decoherence rate is not a rate                                          | proved              | T |
| Invariance alone selects no dynamics                                            | proved              | T |
| Minimal scale usage selects $n = 2$                                             | proved              | T |
| One scalar cannot carry two polarisations                                       | proved              | T |
| Clausius route imports $\hbar$                                                  | proved              | T |
| Scalar source is not conserved                                                  | proved              | T |
| $\delta\mathcal{I} = 0 \Leftrightarrow \chi \equiv 1$                           | proved              | T |
| $\Omega = 0$ does not imply $\chi \equiv 1$                                     | proved              | T |
| Selection principle itself; $\lambda = L_{\mathcal{E}}$                         | stated condition    | D |
| Reversible curvature in graphene                                                | measurement pending | S |

## VIII. CONCLUSION

Information invariance, defined as stationarity of the loop functional, is equivalent on finite charts to vanishing thermodynamic curvature and to exactness — a proved statement with an executable certificate. The experimental signature usually quoted for curvature, a nonzero one-cycle loop area, is proved insufficient; the decision is deferred to a predeclared second-cycle test. The dynamical programme connecting this principle to quantum and gravitational phenomena is stated as conjectures with named obligations, and every obligation is carried to a verdict rather than left open. The emergent-gravity claim is withdrawn on three independent grounds — degree-of-freedom counting, the  $\hbar$ -dependence of the thermodynamic route, and non-conservation of the scalar source. For the quantum step the obligation is discharged with a negative outcome: the proposed energy dynamics generates a damped rotation rather than unitary evolution, and the step survives only as the explicit condition that the cut-even symmetric part cancel at the background. We regard reporting this as more useful than asserting the step.

## REPRODUCIBILITY

All theorems marked “proved” are accompanied by executable certificates over exact rational arithmetic, with pinned SHA-256 digests regenerated by continuous integration on every commit; tampering with any recorded value changes the digest and fails the build. Source, certificates

and the claim ledger: Parveen117/Publications, Science, Bengaluru.  
papers/information-invariance.

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- [1] M. Dabas, “Geometric Completion of Thermodynamic Response,” arXiv:2603.20773 [cond-mat.stat-mech] (2026).
  - [2] Parveen117/Thermodynamics-Reproducibility, campaign T02\_RAMAN\_THERMAL\_HYSTERESIS (2026).
  - [3] Dimensional statements in Sec. V use only  $(M, L, T)$  exponents; the numerical illustration uses a vacuum energy density of order  $10^{-9} \text{ J m}^{-3}$ .
  - [4] Recognition-Kernel Framework, `theorems/foundation/FOOE_NATIVE_EULER_FROM_IOTA_COMPLEX` (PROVED), Zenodo doi:10.5281/zenodo.21760119.
  - [5] Recognition-Kernel Framework, `theorem/41_cut_graded_universal_generator_theorem` (certified, pin 34afc445...), *ibid*.
  - [6] T. Jacobson, Phys. Rev. Lett. **75**, 1260 (1995).
  - [7] L. Diósi, Phys. Lett. A **120**, 377 (1987).
  - [8] R. Penrose, Gen. Rel. Grav. **28**, 581 (1996).
  - [9] M. Aspelmeyer, T. J. Kippenberg, and F. Marquardt, Rev. Mod. Phys. **86**, 1391 (2014).
  - [10] A. Bassi *et al.*, Rev. Mod. Phys. **85**, 471 (2013).